

Animal Welfare Enforcement in the UK: Evaluating Risk-based and Technology-led Approaches

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In 2024, 2.2% of UK farms were inspected and 28.4% of those inspections found non-compliance with animal welfare regulation. This presents a major enforcement bottleneck where legal protections may fail to translate into real welfare improvements. This paper examines the factors that contribute to this enforcement problem and evaluates two potential solutions: a centrally-led risk-based approach to inspection prioritisation and the incorporation of precision livestock farming (PLF) technology into welfare monitoring. It concludes with proposals for implementation and further investigation.

The enforcement problem

The enforcement problem mentioned in this paper refers to the low rate of animal welfare inspections on farms across the United Kingdom. Currently, depending on the estimated number of farms, the average farm can expect an inspection once every 14 to 45 years (Animal Law Foundation, 2025). This paper focuses on on-farm inspections within England, Scotland and Wales, excluding other stages of the supply chain such as transportation and slaughter.

Responsibility for enforcement

In England, Scotland and Wales, responsibility for on-farm welfare enforcement is split between local and national authorities, with inspections typically conducted by local authorities and the Animal and Plant Health Agency (APHA) (APHA, 2016). While the Department for Environment, Food and Rural Affairs (DEFRA) in England and the Scottish and Welsh Governments set enforcement policy, they delegate much of its operationalisation to APHA, which in turn decentralises significant responsibility to local authorities. In practice, APHA often views its role as purely regulatory and advisory while local authorities remain the primary bodies for bringing prosecutions (Animal Law Foundation, 2025).

Enforcement triggers and process

Any farm can be inspected in England, Scotland and Wales, though there is no legal requirement specifying how often (APHA, 2016). In practice, checks are triggered by complaints, history or referrals rather than routine scheduling. Inspectors have power of entry and can act in the farmer's absence. Where unnecessary suffering is suspected, an inspection must follow within 24 hours and should normally be unannounced.

Enforcement in practice: key statistics

According to the Animal Law Foundation's 2025 report, 2.2% of UK farms were inspected in 2024, with 28.4% of those inspections finding non-compliance. Despite this, enforcement action is rare: 55.81% of local authorities in England, Scotland and Wales who identified breaches took no formal action and only 2.2% of non-compliance cases led to prosecution. Where action was taken, notices constituted 85% of all enforcement activity. In England, Scotland and Wales, 65.3% of farm-related complaints resulted in an inspection.

Estimating scale of suffering

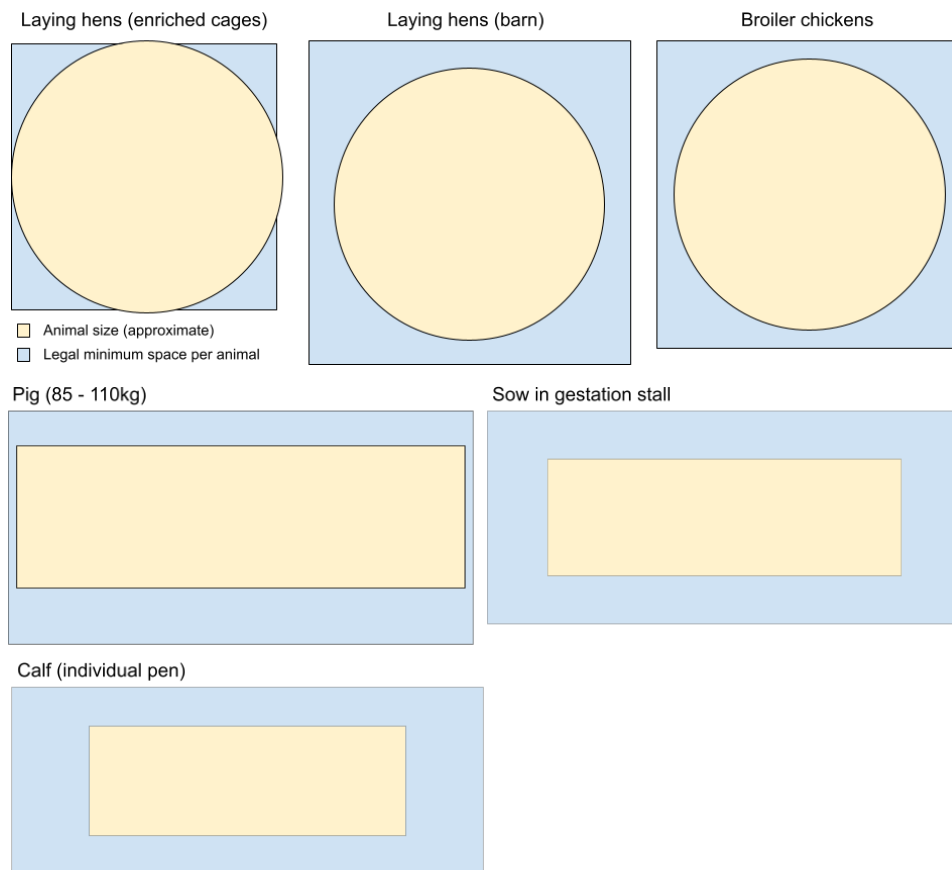
Over one billion land animals were slaughtered in the UK in 2024/2025 (Food Standards Agency, 2025). Including farmed fish pushes the number of animals killed each year to well above one billion (Food Manufacture, 2025), and additional millions are estimated to die on farms before slaughter (The Guardian, 2023). Given that 28.4% of inspected farms are found to be non-compliant with animal welfare requirements, a substantial number of animals are therefore likely to be raised in conditions that risk falling short of legal standards, although the exact number cannot be precisely determined from available data.

It is also important to consider the potential severity of suffering. UK animal welfare regulation is primarily focused on preventing unnecessary suffering rather than promoting positive welfare. The Welfare of Farmed Animals (England) Regulations 2007 set the following baseline duties and minimum space allowances:

- Animals must be inspected at least once a day (if welfare depends on frequent human attention) or at intervals sufficient to avoid suffering
- Animals that appear ill or injured must be cared for appropriately and without delay; if necessary, isolated with suitable accommodation and dry bedding
- Freedom of movement must not be restricted in a way that causes unnecessary suffering or injury
- Continuously restricted animals must get space appropriate to physiological and ethological needs, per good practice

These requirements aim to prevent unnecessary suffering rather than promote positive welfare, so non-compliance means even this basic level of protection is not being met.

Species	Minimum space per animal
Laying hens (enriched cages)	750cm2 total (599cm2 usable space)
Laying hens (non-cage/barn)	~1,111cm2 (9 hens/m2 max)
Laying hens (free-range)	~1,111cm2 indoor + 4m2 outdoor
Broilers (meat chickens)	1000cm2 at slaughter weight
Calves (individual pens, >220kg)	1.5 m2
Pigs (85 – 110kg)	0.65m2
Sows (gestation stall)	2.25m2



The role of assurance schemes

It could be argued that assurance schemes provide some regulation oversight. Assurance schemes were introduced in the UK in the 1990s to indicate to customers how food products were being produced (AHDB, 2025). In exchange for following specific welfare standards, farms are allowed to use scheme labels on their packaging. These schemes conduct inspections to ensure their own scheme membership rules are being followed. If a farm is found to be non-compliant, it can be suspended or lose its membership entirely. Although comprehensive membership data is not publicly available, assurance schemes are a dominant force as major UK supermarkets require scheme membership as a condition of sale (van der Ven et al., 2022). This has created a two-tier regulation system where the government sets the legal baseline and private assurance schemes act as more frequent auditors.

The existence of assurance schemes partly explains low government inspection rates: their presence creates an impression of oversight that can make the 2.2% figure seem less alarming. The non-compliance rate may therefore be the more informative measure. Government inspections found non-compliance in 28.4% of farms. Unfortunately, assurance schemes rarely publish full inspection details so their actual compliance rates are unknown (DEFRA, 2012). However, assurance schemes likely cover most UK agricultural production (Red Tractor, 2026; van der Ven et al., 2022), so most government-inspected farms are probably scheme members. Assuming inspected farms are broadly representative, the 28.4% figure can therefore be read as a rough indication of non-compliance despite assurance scheme oversight. This paper does not set a benchmark for an acceptable rate or compare it with other regulated sectors, so it

cannot say whether 28.4% is high relative to other regulated sectors. The judgement that it is concerning rests on the view that the rate should be close to zero, since these are breaches of legal requirements.

Furthermore, there is a fundamental gap between the enforcement capacity of assurance schemes and the law. Assurance schemes are market-based, meaning they can strip a farm of a label or block it from selling its products in a supermarket but they cannot issue statutory notices, fines or bring criminal prosecutions. They therefore cannot replace state enforcement as they have no power to legally compel compliance

Key factors contributing to the enforcement problem

It could be argued that fragmented responsibility contributes to the enforcement problem. Farm animal welfare enforcement is distributed across multiple authorities and shifts as animals move through the production chain. On-farm inspections are undertaken by local authorities as well as relevant government departments and agencies. With the exception of Northern Ireland, which operates a more centralised model, the enforcement structures are complex and involve multiple authorities (Animal Law Foundation, 2025).

In some cases, public bodies' stated responsibilities differ from their legal duties. For example, the Scottish and Welsh governments stated they did not enforce farmed animal welfare legislation, rather it is enforced by local authorities and in the case of the Scottish Government, APHA. Yet APHA stated they perform inspections only as a regulatory rather than enforcement body, and therefore they do not bring or refer prosecutions (Animal Law Foundation, 2025). This goes against APHA's regulatory and compliance policy that states it may refer matters for investigation or prosecution (APHA, 2017). The Scottish Government also claimed that APHA enforces farmed animal welfare legislation in Scotland, rather than the Scottish Government (Animal Law Foundation, 2025). These inconsistent accounts suggest a fragmented system where responsibility is unclear even to the authorities involved. Fragmented responsibility makes it difficult to monitor enforcement activity or identify where failures occur and creates conditions where authorities may defer responsibility to one another.

Another factor that may contribute to the enforcement problem is resource constraints. In Scotland there is one inspector for every 1327 farms; in England, one for every 892 and in Wales one for every 657. Northern Ireland with one inspector per 79 farms is an outlier, possibly reflecting its more centralised enforcement model. Across the UK as a whole, the ratio stands at one inspector per 490 farms (Animal Law Foundation, 2025).

Funding is likewise a significant constraint. A May 2025 report by the Chartered Trading Standards Institute found that 84% of local authorities considered cost a barrier to enforcement (Chartered Trading Standards Institute, 2025). Individual cases can exceed £200,000 when court delays, legal proceedings and the cost of caring for seized animals are factored in. More broadly, local authorities report insufficient dedicated funding for animal welfare enforcement, covering staffing, legal capacity and physical infrastructure.

It is possible that the current complaint-led, reactive approach lacks an effective deterrent. This sort of model allows certain farm types - such as new businesses, remote premises or indoor facilities - to avoid

common inspection triggers. Consequently, welfare issues on these farms may remain undetected for significantly longer.

The slow pace of court proceedings further weakens the deterrent effect of the current system. Cited by 39.4% of surveyed officers as a key obstacle to enforcement (Chartered Trading Standards Institute, 2025), it can make prosecution seem too time-consuming an option for under-resourced local authorities.

Institutional responses and policy developments

Following the release of Animal Law Foundation's reports, several bodies published independent evaluations of the enforcement system:

- February 2025: The Animal Sentience Committee issued a report noting that legislative compliance and enforcement were insufficient (Animal Sentience Committee, 2025)
- May 2025: The Chartered Trading Standards Institute identified financial costs and legal delays as primary barriers to effective enforcement (Chartered Trading Standards Institute, 2025)
- June 2025: findings were presented to the All-Party Parliamentary Group for Animal Welfare (APGAW, 2025)
- July 2025: A House of Lords scrutiny report raised concerns regarding non-compliance in the poultry sector and called for greater clarity on future enforcement guarantees (House of Lords Secondary Legislation Scrutiny Committee, 2025)

In December 2025, the government published its Animal Welfare Strategy for England (DEFRA, 2025). The document acknowledged the importance of enforcement in maintaining welfare standards; however, it did not propose new enforcement funding, structural reorganisation of regulatory bodies or changes to existing enforcement protocols. Therefore, despite repeated independent concerns, the government's response has so far left the resourcing and structure of enforcement unchanged.

The enforcement problem is not solely a question of resources - fragmented responsibility and unclear accountability suggest that additional funding alone would not resolve the underlying structural issues. The remainder of this paper evaluates two approaches, a centralised risk-based inspection framework and the incorporation of PLF technology into welfare monitoring; both aim to make better use of existing capacity rather than increase it. This does not mean more resources are unnecessary; the evidence above suggests they are needed. However, better prioritisation is worth pursuing either way. Under-resourced enforcement that targets the right farms is preferable to under-resourced enforcement that does not, and if funding does increase then a risk-based model helps ensure it is used effectively.

Solution one: a centrally-led risk-based approach

What is a risk-based approach to the enforcement problem?

A risk-based approach to the enforcement problem would allocate inspections according to the likelihood and severity of non-compliance so that limited capacity goes where it is most needed. It differs from the current, reactive system where inspections are largely triggered by complaints (APHA, 2016). A risk-based approach is preventative because it aims to find problems before they are reported.

What risk-based approach should we aim for?

An effective risk-based approach should find more non-compliance than the current complaint-led model for the same number of inspections, since capacity is limited. This includes cases that would never be reported. It should also be practical to implement, fair, and leave no farm entirely uninspected.

How would the new approach work?

The goal is to move from a complaint-led inspection schedule to one based on multiple factors that predict non-compliance so that authorities can direct limited resources where they are most needed. There are several ways to identify these factors and decide how much each should count. Examples include:

- Statistical analysis of past inspection data (e.g. regression) to find which farm characteristics are most strongly linked to breaches.
- Expert judgement where senior inspectors and veterinary experts rank known warning signs by importance (e.g. a history of incomplete veterinary records may be judged as a stronger indicator than farm size).
- Literature review, drawing on published research on risk factors for poor welfare.
- A combination, such as experts adjusting the weights from a statistical model.

Whichever method is used, each farm would be given a risk score and ranked. Farms could then be grouped into three risk bands:

- High risk: the highest-scoring farms that inspection capacity allows (for example, the top 10 -20%). Inspected every 6 - 12 months, with unannounced visits to observe normal operating conditions.
- Medium risk: elevated but not critical risk. Inspected every 12 - 24 months, using a mix of announced and unannounced visits.
- Low risk: the lowest-scoring farms. Inspected every 2 - 3 years to maintain oversight.

In terms of who might lead this approach, the Animal and Plant Health Agency (APHA) is one plausible lead. It already collects data on welfare complaints, inspection activity and non-compliance findings (House of Commons, 2025), and its 2021-2026 science strategy names data science as a priority (APHA, 2021). It also works with local authorities and employs veterinary experts, so it could coordinate both the data analysis and the use of expert judgement.

Safeguards

To stop the risk model creating blind spots or replacing professional judgement:

- Random inspections across all bands (a small fixed share of capacity) would test the accuracy of the risk assessment and help identify new predictors of non-compliance rather than only reinforcing the factors already used.
- Capacity would be allocated in this order:
 1. Credible complaints, which trigger an inspection that overrides the ranked list
 2. Random inspections

3. The ranked risk list

- Any deviation from the ranked list must be documented with reasons so inspectors' judgement is recorded and can be reviewed.

How would it realistically be introduced?

A pilot programme could be introduced across a limited number of local authorities with varying geography and farm density to test whether the approach works in different conditions. The first step would be to develop the risk assessment using existing enforcement and farm data, then set the risk bands and the inspection procedure for each (for example, inspection frequency and whether visits are announced or unannounced). The selected local authorities would then use these risk scores to plan inspections. Inspection outcomes would be recorded and fed back to the central team to refine the risk assessment. In order to evaluate the effectiveness of the pilot approach, the results would be compared with a group still using complaint-led inspections.

How will we know if the approach is successful?

1. Detection: with the same number of inspections, the pilot area should find more non-compliance than a matched control area. A rising non-compliance rate among inspected farms should not be read as declining standards, rather it is the expected result of concentrating inspections where breaches are most likely. If the rate rises more in the pilot area than in the control area, this is consistent with the model finding breaches that were previously undetected.
2. Sustainable workload: inspector workload in the pilot area should be no higher than in the control area, or at least be reported as manageable by officers. Even if high-risk flags exceed capacity, the model may still outperform complaint-led inspection, but persistent backlogs, rushed inspections or displaced complaint responses would weaken it
3. Efficiency (cost per breach): the cost of finding each breach should be lower than in the control area. If the system is working, the same spend should find more breaches, because inspections are directed at the farms most likely to be non-compliant.
4. Predictive accuracy: this compares risk categories within pilot, not pilot vs control (see point 1). The risk assessment is reliable if "high-risk" farms have a clearly higher breach rate (threshold to be determined) than "low-risk" farms measured using random inspections within each band. If not, the risk assessment needs revising before it is scaled up.

What failures should be anticipated?

- Inaccurate risk assessment/model: if the available data for a model are insufficient or poor quality, or perhaps an expert-led approach misses key factors, the risk assessment may fail to predict non-compliance reliably. In that case, farm categorisation and the resulting inspection priorities are unlikely to improve detection. Misclassification could also occur (e.g. high-risk farms categorised as low-risk) resulting in serious harm going undetected.
- Implementation cost: the approach would require additional resources, such as a central analytical or coordination team. In the short term, improved targeting may uncover more non-compliance, increasing enforcement costs. But in the longer term, depending on penalties being

credible, more effective detection may act as a deterrent and reduce the number of breaches thereby decreasing enforcement costs.

- Fragmented responsibility: since fragmented responsibility is a contributing factor to the enforcement problem, the approach would be less effective without clearly defined roles. Unclear coordination between authorities or unclear responsibility within them risks reproducing existing governance problems.
- Slow court proceedings: the approach does not address delays in prosecution. Improved detection may add pressure to an already slow court system.

Does this solution improve on the current fragmented system?

As mentioned previously, fragmentation seems to contribute to the enforcement problem, so a system that increases consistency between local authorities (clear, transparent and standardised, and easier to monitor and improve) is preferable.

This approach should reduce inconsistency in how inspections are prioritised, since risk scores would be calculated for all farms using the same standardised risk assessment. Roles would need to be clearly defined. The central regulator would develop the assessment, regularly review data and evidence to identify risk factors, and return the risk list to local authorities. Local authorities would record and report enforcement activity in their area, including complaint and prosecution history, and inspect in order of priority. The central regulator could still conduct inspections itself, but primary responsibility would remain with local authorities.

Lessons from other government risk-based approaches

Previous government risk-based approaches offer lessons for implementation.

The Food Standards Agency (FSA) pilot programme (January 2021 to March 2022) aimed to improve food safety compliance by targeting the areas posing the greatest risk (FSA, 2023). It reviewed the existing framework and introduced a business risk scoring system and decision matrix, forming the Food Standards Delivery Model (FSDM). Seven local authorities were trained to use it and reported that it assessed risk more accurately and fitted easily into existing practice. The pilot's key strength was strong communication, training and ongoing support between central government and local authorities.

Criticised approaches also offer lessons. The post-Brexit Border Target Operating Model has been challenged because its new risk categories were considered too broad (House of Commons Environment, Food and Rural Affairs Committee, 2025). Critics argue that, instead of targeting more precisely it simply increased checks across many different groups. For animal welfare, this suggests the aim should be more precise targeting, not a wider net that burdens everyone without a clear benefit.

Could it be gamed?

Farms could try to game the system in two main ways. First, they could avoid displaying the indicators that best predict non-compliance while still breaching welfare regulations. This depends on how easily indicators can be hidden and it also depends on farms knowing which indicators are used. Second, if farms learn their risk band then low-risk farms in particular may have less incentive to comply as they would

expect infrequent inspections. Keeping indicators and bands undisclosed would limit both. Random inspections across all bands and complaint-triggered overrides reduce the remaining effect but do not eliminate it, and their impact depends on the proportion of random checks.

Overall assessment

I think this is a promising risk-based option given limited resources. It offers a standardised way of prioritising inspections that in theory should improve detection of non-compliance. Its success would depend on frequent communication between local authorities and the central regulator and potentially a data sharing platform. Inspection capacity should be allocated to complaints first, then random inspections, then the risk list, with local intelligence used to justify documented deviations. A pilot would test feasibility and gather feedback before wider rollout. If welfare outcomes do not improve even when the system works as intended, the bottlenecks will need to be reconsidered.

Solution two: PLF as continuous monitoring PLF welfare monitoring

If precision livestock farming (PLF) technology becomes more widely adopted on farms, and if the data it generates can be used to assess animal welfare, there may be an opportunity to incorporate it into regulatory oversight. This section examines whether both conditions hold and what it would mean in practice.

What is PLF?

PLF uses sensors, cameras and microphones to monitor animals and their environment. These devices record movement, sound, temperature and location which a system uses to trigger a response. This could be an automated action, such as adjusting lighting or sending an alert to the farmer, for example if an animal is showing early signs of lameness (Carolin et al, 2017).

A key aim of PLF is to keep track of individual animals as farms grow. On large farms it is impractical for staff to monitor every animal at all times, so PLF can fill this gap. These tools can also save labour and may raise productivity by identifying the specific factors that lead to the greatest yields (Bernabucci et al., 2025).

Current adoption trajectory

There are no up-to-date official statistics on PLF adoption in the UK. Survey data from England in 2019 shows that uptake varies by technology and sector (DEFRA, 2020). For example, 64% of pig and poultry farms used automated weighing to monitor growth, while fewer than 10% of grazing livestock farms used pasture sensors.

Productivity was the main reason given for adopting PLF. 78% of surveyed farms cited productivity and performance, while only half cited animal health and welfare improvement. Barriers include the perceived cost-effectiveness of the tools and whether they suit the farm type (DEFRA, 2020).

Government schemes such as the Farming Equipment and Technology Fund and the Farming Innovation Programme may help fund PLF, though they are not specific to it (DEFRA, 2026).

The trajectory is therefore uncertain. Adoption is uneven and driven mainly by productivity. Arguably, it

could increase if costs fall, productivity benefits are better demonstrated and funding becomes available, but the 2019 data cannot confirm this.

Could PLF farming technology be used to assess animal welfare?

This table shows the types of technologies used to measure different welfare indicators related to pig production (Gómez et al., 2021).

Welfare indicator	Inputs	Measured traits
Activity level	Cameras, accelerometers (wearables), thermal cameras, RFID*	General motion (active/inactive state), tracking location, postural states (standing and lying), and transition between states.
Feeding & drinking	RFID, cameras, accelerometers, load cells, microphones	Intake volume, frequency and duration of feeding/drinking; nursing/suckling behaviour or social competition at stations.
Social & specific behaviours	Cameras, accelerometers, microphones	Aggression, social grouping and species-specific stress behaviours (e.g., tail-biting, mounting)
Physical condition	Load cells, 3D cameras, microphones, thermal cameras	Gait and weight distribution (lameness detection), body weight/condition scoring, vocalisations (coughs/distress) and surface body temperature
Health status	Cameras, accelerometers, infrared thermography, RFID, microphones	Early detection of disease (e.g., respiratory issues, diarrhoea or mastitis), hoof lesions and thermal
		changes indicating fever or inflammation.

* Radio frequency identification: wireless technology that identifies, tracks and monitors individual animals in real time.

How PLF can be technically used to ensure compliance with animal welfare regulation

From a technical perspective, PLF can plausibly be used to gather, process and interpret data on animal welfare in farm settings. One study found there are over 80 PLF tools commercially available that can potentially be used to assess animal welfare indicators on pig farms (Gómez et al., 2021). PLF tools can target specific areas of welfare regulation. For example, overhead cameras can be used to measure pen occupancy to determine whether stocking density regulation is adhered to. Similarly, PLF could be used to help verify the UK law banning the confinement of calves over eight weeks old to individual stalls and pens - accelerometers can detect play and grooming between calves, and the absence of these behaviours could indicate non-compliance and prompt an inspection. Cameras can also be used to check for illegal handling of animals (Gómez et al., 2021).

However, the existence of these technologies does not settle how PLF data should be incorporated into

regulatory systems and what type of systems would be best suited to PLF integration. Several design considerations would need to be addressed around procedural fairness, uneven applicability across farms and data management for PLF data to effectively support enforcement.

To ensure compliance with UK standards of procedural fairness, it may be preferable for PLF data to function as a regulatory trigger rather than a prosecution tool. For example, rather than outputting a welfare score (which raises black box concerns) systems could output a summary of raw facts (e.g., animal #402 spent 40% less time standing than the previous seven-day average) flagging where inspections should be undertaken. The inspector would retain expert witness status while leveraging the continuous monitoring capacity of PLF technology. This complementary role is also important given that PLF is harder to implement on extensive grazing farms (sheep and cattle) than in intensive indoor settings (chickens and pigs), meaning it cannot easily be applied uniformly across the sector and should therefore not completely replace existing inspection efforts (Morgan-Davies et al., 2017). A practical further consideration is data volume: rather than a constant stream, periodic summary reports would be more manageable for inspectors, preserving the benefits of continuous monitoring without overwhelming existing workflows.

Governance models

It seems plausible that PLF tools can technically capture meaningful data related to animal welfare. But how is this data managed and who should see it? There are four key governance models for using PLF to enforce animal welfare regulation.

The first is the farmer-managed feedback loop: PLF systems are chosen by farms and data is fed back to farmers who act upon welfare alerts. This would offer the lowest level of transparency as only farmers would know if welfare is being handled or not. It would be easiest in terms of implementation as it only requires one level of welfare handling rather than sharing the role of regulation with multiple groups. However, enforcement capacity would be low: whether a welfare alert is acted upon is entirely at the farmer's discretion. Although it is the easiest to implement and least likely to be opposed by farmers, it offers the weakest guarantee of compliance and transparency.

The second model is company integration, where welfare data is shared with the PLF provider, who monitors and enforces compliance. Implementation would be relatively easy, since companies already manage the data infrastructure. Enforcement would be set by individual companies and could range from financial penalties to softer incentives like benchmarking or gamification. Under this approach, companies would have weak commercial incentives to impose strict welfare safeguards, especially if these safeguards make products less appealing. Transparency would remain limited and enforcement would likely vary between providers. Overall, this model offers more oversight than the farmer-managed approach, but it still relies on private companies to uphold legal standards. At best it could complement existing regulation, but it is unlikely to ensure strong or consistent compliance on its own.

The third option is farm assurance scheme integration. Under this model, farms would be required to share relevant PLF data directly with the assurance schemes they belong to. This creates a data-led audit system where schemes can periodically review digital records alongside physical inspections. In practice, this would require schemes to accredit specific PLF systems that provide welfare-relevant data and

demonstrate high accuracy. To incentivise farmers to share system-generated data, they could be rewarded with “earned recognition”, resulting in fewer physical inspections (FSA, 2024). In theory, this model could be one of the strongest for accountability since schemes state that their aim is to uphold welfare standards. However, risks remain: if schemes both approve PLF systems and judge compliance using the data those systems generate, conflicts of interest may arise. There is also a risk that accreditation decisions could be influenced by commercial partnerships between tech companies and schemes rather than being driven purely by animal welfare outcomes. Overall, using PLF data in assurance schemes could help reward compliant farms with fewer inspections while flagging those that need attention. However, it seems that government-created safeguards may be required to ensure that animal welfare is not sacrificed for profit.

The final model is direct government integration. Under this approach, any PLF system collecting welfare-relevant data would be required to make that information accessible for regulatory oversight. For example, PLF providers could enable access for central regulators who would audit the information and delegate enforcement to local authorities. Implementation would require legislation mandating government access to the records held by PLF companies, alongside a standardisation of welfare metrics. These standardised reports could be generated by the tech companies or produced by regulators from raw data. However, direct company-to-government transmission may raise serious concerns for farmers regarding surveillance and data ownership. A more workable approach might be to require PLF systems to generate a standardised welfare report that goes to the farmer first, who then submits it as part of their legal compliance. There is also a practical risk: if government oversight is perceived as punitive rather than supportive, farmers may be discouraged from investing in PLF products that generate welfare data. Overall, poor implementation of this approach could create resistance without improving outcomes for farmed animals. Therefore, for this model to succeed, it must be carefully designed to ensure credibility, fairness and genuine welfare improvement

Comparative assessment: assurance scheme vs government integration Enforcement and institutional integrity

Government integration would likely provide more direct and legally grounded enforcement because those responsible for interpreting compliance also have the authority to sanction breaches. In contrast, assurance schemes introduce an intermediary layer whose effectiveness must itself be monitored.

Assurance schemes face possible conflicts of interest, especially if they both approve PLF systems and evaluate compliance. Schemes are also funded by their members and may have commercial ties to PLF companies which could make them reluctant to penalise farms or suppliers. Government enforcement is likewise open to lobbying from farmers, PLF companies and other parties. However, government regulators have less clear commercial stakes in whether farms or PLF companies succeed and their decisions can be challenged through public accountability routes such as judicial review, so they may be better placed to resist this pressure.

Scale

Not all farms are members of assurance schemes, whereas all farms are subject to government enforcement. Government integration would therefore in principle reach more farms, helping more of

them meet at least the minimum animal welfare standards set out in law.

However, government integration would only enforce the legal minimum whereas assurance schemes may be better positioned to secure higher levels of welfare: whilst some schemes begin with the legal minimum as a baseline, some, such as RSPCA Assured and organic standards, go beyond statutory requirements (Bartlett et al., 2023). Schemes that integrate PLF could therefore embed welfare standards that exceed the legal minimum rather than focusing solely on baseline compliance.

Feasibility and political durability

Assurance schemes may allow for quicker implementation since some schemes already have membership portals and digital auditing processes in place (Red Tractor, 2025). Adapting these existing systems to include PLF data would be a smaller step than starting from scratch. Furthermore, government integration would likely require significant legislative reform and a much larger administrative capacity to build and manage a national data network.

Both models need to be acceptable and durable. A model is more likely to be feasible if it avoids significant backlash from farmers, the public and the industry; an optimised welfare solution is of little value if political pressure leads to it being scrapped so it must be designed to survive political and industry shifts. While this is a challenge for both approaches, assurance schemes are arguably more viable because they are less bureaucratic: their smaller scale may allow them to adapt quickly, troubleshoot problems and respond to industry feedback more effectively than a government department.

Final judgement

In the short term, assurance schemes may be best placed to pilot PLF data as welfare indicators. Because they already run farm audits and have systems for collecting member data, they may be able to add PLF data to existing processes and fix problems more quickly than government departments.

In the long term, if assurance scheme pilots prove successful, the government could move towards full integration. This would require farms with PLF systems to submit key welfare statistics directly to the regulator. This participation could be incentivised through earned recognition (rewarding farms with fewer physical inspections unless the data reveals patterns of non-compliance). Ultimately, government integration may be the long-term ideal due to its stronger enforcement capability and its ability to reach farms that are not part of assurance schemes.

Discussion

Responsibility for farmed animal welfare regulation in Great Britain is shared between APHA and local authorities, with inspections often triggered by complaints. Assurance schemes provide additional oversight, but can only sanction through loss of membership rather than legal prosecution. APHA has encouraged authorities to prioritise local intelligence over centrally set inspection schedules (DEFRA, 2022). Inspection and non-compliance rates have since remained stagnant (Animal Law Foundation, 2025), suggesting this reactive approach has not improved outcomes.

A government-led enforcement strategy could improve transparency (centralised access to data across all farms), consistency (same standards applied to all farms) and legal accountability (non-compliance has

statutory consequences).

An effective risk-based model would require a designated central regulator to create and manage a standardised risk-assessment framework. It would also likely require data-sharing infrastructure where local authorities can submit enforcement data for central evaluation to update risk models. Central regulators would determine farm risk levels and communicate them to local authorities who retain responsibility for on-the-ground enforcement. Before national rollout, a pilot across a mixture of local authorities could test feasibility and gather inspector feedback.

It should be noted that this proposed risk-based approach is an efficiency rather than structural reform. Its value depends on how inefficient the current system is at targeting inspections. If the approach identifies non-compliance more effectively but welfare outcomes do not improve in the pilot, this would suggest that detection is not the only constraint. The bottleneck may lie in what follows detection, such as penalties and prosecution of breaches.

In terms of PLF as continuous monitoring, integration with assurance schemes may be more feasible in the short term. Incorporating PLF systems into statutory regulation would likely require coordination across multiple government departments as well as the development of new legislation and regulatory frameworks. In contrast, private assurance schemes can typically update their standards more rapidly. Because PLF-based welfare monitoring is still developing, these schemes may provide a more suitable environment for early experimentation and refinement.

Meanwhile, the government could introduce legislation around PLF. Firstly, the government could regulate which PLF products are allowed on the market. Systems could be required to be tested in commercial farm conditions to prove they do not harm animals. Secondly, assurance schemes could be required to be transparent about how they interpret raw data as welfare indicators. By making their methods public, the government and the general public can check and challenge whether these schemes are actually protecting animals or simply “humane washing” the data. Finally, there could be rules to prevent assurance schemes from promoting PLF products based solely on how lucrative it is for the farm. Any system they approve could be required to show that it improves the lives of animals, not just the efficiency of the business.

Conclusion

The enforcement problem has structural issues that additional resources alone would likely not resolve. Fragmented responsibility, a reactive inspection model and limited transparency contribute to a system in which on-farm adherence to legal welfare requirements are rarely checked and are breached in approximately a third of cases.

A centrally-led risk-based approach offers an actionable path forward, directing existing inspection capacity to higher-risk farms, though its gains will be limited if what follows detection (penalties, prosecution, correction of breaches) remains a bottleneck. In the longer term, PLF technology could supplement one-off inspection visits with continuous monitoring, but only if its data is managed transparently and in the interests of animal welfare rather than commercial efficiency.

Neither solution is a quick fix and both depend on clearer responsibilities between enforcement bodies

and PLF also needs standardised welfare metrics and rules on data access. Both should be tested through pilots before national rollout.

Bibliography

Agriculture and Horticulture Development Board (2025) UK Farm Assurance Review

All-Party Parliamentary Group on Animal Welfare (2025) APGAW Roundtable on Enforcement

Animal and Plant Health Agency (2016) Animal welfare on farms inspection

Animal and Plant Health Agency (2017) Regulatory and Compliance Policy

Animal and Plant Health Agency (2021) APHA Science Strategy 2021 to 2026

Animal Law Foundation (2025), The Enforcement Problem: 2024 Data Report

Animal Sentience Committee (2025) Report on the Due Regard to Animal Welfare: Legislative Compliance and Enforcement

Bartlett, H., Balmford, A., Holmes, M.A. and Wood, J.L.N. (2023) Advancing the Quantitative Characterization of Farm Animal Welfare, Proceedings of the Royal Society B: Biological Sciences

Bernabucci, G. et al. (2025) Precision livestock farming: an overview on the application in extensive systems, Italian Journal of Animal Science

Carolin, J., Yasotha, A. and Sivakumar, T., 2017. Precision livestock farming: An overview. Indian Journal of Animal Production and Management

Chartered Trading Standards Institute (2025) Tackling Animal Harm

Department for Environment, Food & Rural Affairs (2020) Farm Practices Survey Autumn 2019: England

Department for Environment, Food & Rural Affairs (2022) Animal Health and Welfare Framework

Department for Environment, Food & Rural Affairs (2025) Animal Welfare Strategy for England

Department for Environment, Food & Rural Affairs (2026) Farming Equipment and Technology Fund 2026: Guidance Now Available, Defra Farming Blog

Department for Environment, Food and Rural Affairs (2012) Farm Animal Welfare: Health and Disease

Department for Environment, Food and Rural Affairs and Animal and Plant Health Agency (2017) Animal Welfare Inspections on Farm

Food Manufacture (2025) Farmed fish welfare put on the agenda in Scotland, Food Manufacture

Food Standards Agency (2023) Evaluation of Food Standards Pilot: Introduction

Food Standards Agency (2024) Earned Recognition: Approved Assurance Schemes.

Food Standards Agency (2025) Animal Welfare Report 2024/25

Gómez Y, Stygar AH, Boumans IJMM, Bokkers EAM, Pedersen LJ, Niemi JK, Pastell M, Manteca X and Llonch P (2021) A Systematic Review on Validated Precision Livestock Farming Technologies for Pig Production and Its Potential to Assess Animal Welfare

House of Commons (2025) Written Question: Animal Welfare: Crime (UIN 73917)

House of Commons Environment, Food and Rural Affairs Committee (2025) UK-EU Trade: Towards a Resilient Border Strategy. Fourth Report of Session 2024-25

House of Lords Secondary Legislation Scrutiny Committee (2025) Thirty-Second Report of Session 2024-25

Kleen, J.L. and Guatteo, R. (2023) Precision Livestock Farming: What Does It Contain and What Are the Perspectives?, Animals

Morgan-Davies, C., Lambe, N., Wishart, H. and Waterhouse, A. (2017) Impacts of Using a Precision Livestock System Targeted Approach in Mountain Sheep Flocks, Livestock Science

Red Tractor (2025) New Portal Features: Making Life on the Farm Easier, Red Tractor

The Guardian (2023) Premature death of 80m chickens raises concerns over UK's fast-growing breeds, The Guardian

The Welfare of Farmed Animals (England) Regulations 2007

van der Ven, C. et al (2022) Core Environmental Standards for UK Imported Agri-Food Products: Options for Pesticide and Fertiliser Use